

BioCoLoop: Collaborative Evidence-Guided Research Framework

OUTER RESEARCH LOOP

Fixed research language model $\mathcal{S}$

Aggregated development evidence $\mathcal{E}_t$
(configuration, dev score, decision)

① **Propose**



$\mathcal{S}$ generates new hypothesis / experiment / expected effect / **design_id**

② **Instantiate**



executable design

③ **Train + evaluate**



④ **Retain / revise**



Candidate model design and training hyperparameters

Development evidence $\mathcal{E}_t$
(one card per trial)

Selected (highest dev score)

INNER COLLABORATIVE TRAINING



Shared coordinator (aggregation/selection service)

broadcast model parameters | aggregate local updates

model parameters

local updates

Laboratory 1



private data



local train + dev



model update

Laboratory 2



private data



local train + dev



model update

Laboratory K



private data



local train + dev



model update

OUTPUTS

Task-specific fitted models



Reusable computational solutions

📄 Designs

`</>` Programs

🛡️ Policies

BIOLOGICAL VALIDATION SETTINGS



Drug-target interaction

Input: compound+protein seq
Output: interaction probability



Proteomic efficacy

Input: proteome profile + drug
Output: drug efficacy



Cell perturbations

Input: expression response profile
Output: perturbation identity